

Month 1: Purple Teaming Batch 01

Network Security & Protocol Analysis

Week 1: Network Security Lab Foundations

Instructor: **Muhammad Saad**

What is Virtualization ?

Virtualization means creating a *virtual (not real but software-based)* version of something — like a computer — inside a physical computer.

If you have **one physical computer**, virtualization lets you run **multiple computers (virtual machines)** inside it.

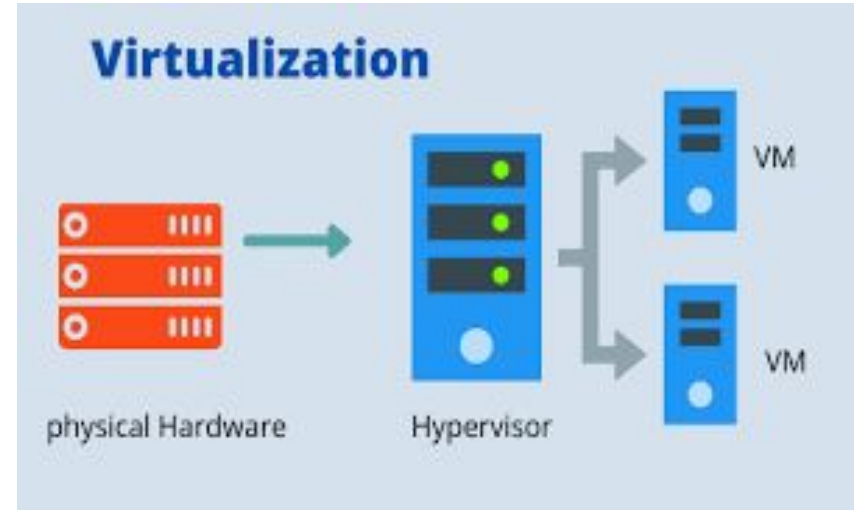
What is Hypervisor ?

A **Hypervisor** is the software that makes virtualization possible.

A **Hypervisor** is a manager that:

- Sits between the **physical computer (hardware)**
- And the **virtual machines**

It connects and controls everything.



Internal Network (Private Network)

An **Internal Network** is a company's **own private network** used inside the organization.

It is a network that:

- Works **inside a building or company**
- Is **not directly accessible from the internet**
- Uses **private IP addresses**

Examples of systems inside an internal network:

- **Database servers**
- **Main company servers**
- **Employee computers**

Internal networks usually use **private IP addresses** like, These IP addresses **cannot be accessed directly from the internet**.

External Network (Public Network)

An **External Network** is the network that is **accessible from outside**, usually through the internet.

It is used so:

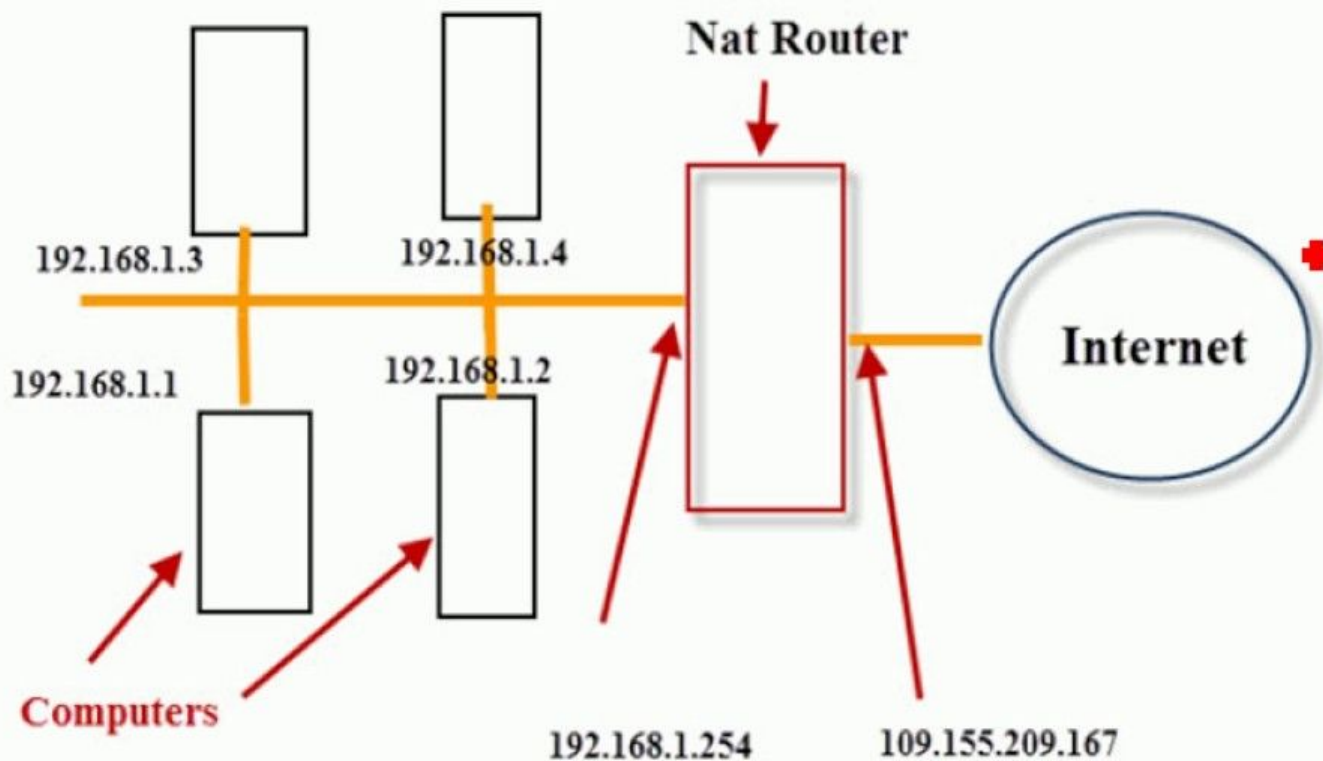
- Customers can access company services
- Users can visit websites
- Public services are reachable

It uses **public IP addresses**, which:

- Are unique worldwide
- Can be accessed over the internet

Examples:

- Company website
- Online services



Home Network Connected to Internet and IP Addresses

NAT – Network Address Translation

- Converts **private IP** → **public IP**
- So devices inside your private network can access the internet
- Keeps track of connections using **port numbers**

Virtual Machines (VMs)

- Usually get **private IP addresses**
- Example: 10.0.2.15 (in VirtualBox NAT mode)
- They are inside a virtual internal network

Physical Machine

- Also usually has a **private IP**: Example: 192.168.1.5
- It is connected to your router.

Router

- Has:
 - One **private IP** (inside network)
 - One **public IP** (given by ISP)

This is the device that actually performs **NAT**

TCP Three-Way Handshake

TCP (Transmission Control Protocol) is a communication protocol that is **connection-oriented, stateful & ensures reliable data delivery**.

Before sending real data, TCP must first **establish a connection** between two devices.

Step 1: SYN

The client sends a **SYN (synchronize)** message.

It basically says: Hello, I want to start communication.

Step 2: SYN-ACK

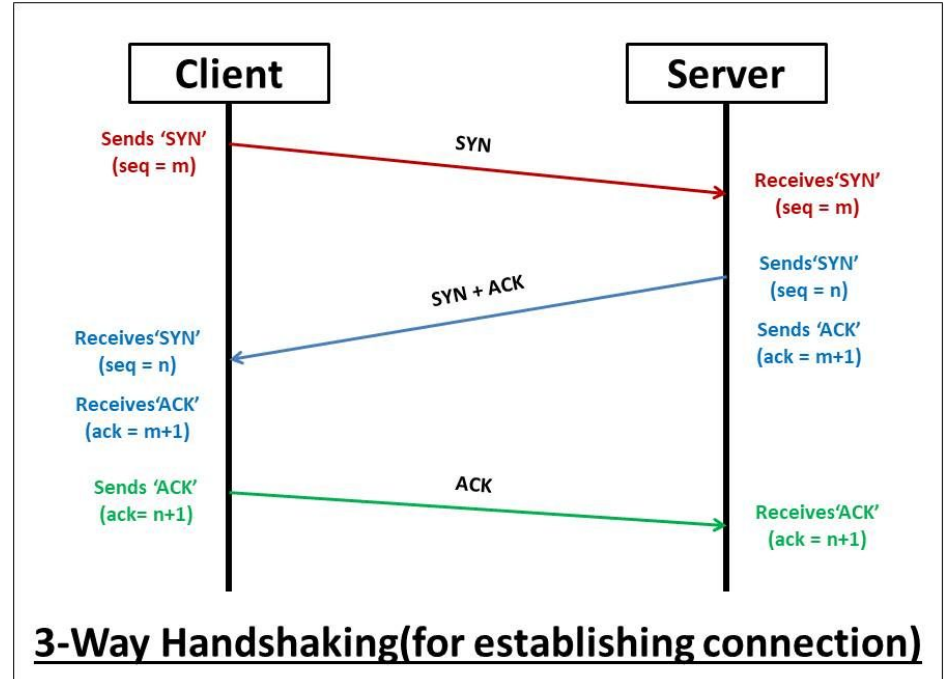
The server replies with **SYN - ACK (acknowledgment)**.

It means: I received your request, and I'm ready to communicate.

Step 3: ACK

The client sends back **ACK**.

It means: Connection confirmed. Let's start sending data.



Best of Luck for the Week 1 Tasks

Instructor: **Muhammad Saad**